SOMNUS

A Descent into Sleep

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I. Introduction

This section introduces the project by covering existing problems and possible opportunities of sleep education.

Introduction

The Problem

Sleep education is often presented through...



Charts



Diagrams



Heavy Text

Introduction

Additionally...

- Wellness content online often **oversimplifies sleep science**
- Many physical changes are **difficult to conceptualize spatially**

Introduction

As a Result...

They often fail to convey the **dynamic, layered nature** of sleep architecture.

Complex neurological and physiological processes remain **abstract and difficult to understand.**

Introduction

The Opportunity

To transform sleep education into an **interactive experience** that allows users to feel and explore how sleep deepens over time.

II. Project Overview

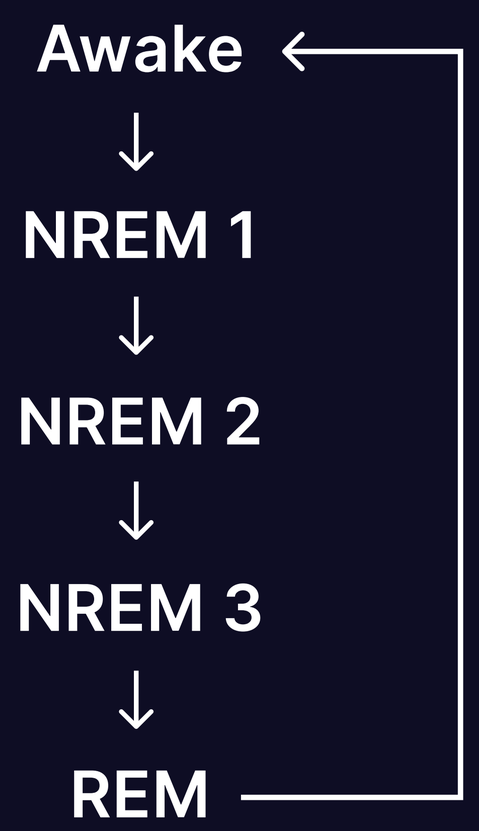
This section covers what the project is, the project goals, intended audiences, environments, and devices.

Project Overview

The Project

Design an interactive website that would...

transform the sleep cycle into an **immersive, scroll-based experience** where users descend through stages of the sleep cycle.



Project Overview

Project Goal

Goal

To help users understand complex neuroscience processes and **visualize the four major stages of sleep.**

How?

Through **motion, depth, and exploration** rather than static information.

Project Overview

Target Audience

Primary

High school and college students learning health sciences.

Secondary

Educators seeking engaging tools to support teaching. General learners interested in sleep and brain sciences.

Project Overview

Target Environment

Classrooms, computer labs, libraries, and individual at-home learning environments where users can focus and **interact at their own pace.**

Project Overview

Primary Devices



Desktop



Laptop

Project Overview

Possible Use Cases

Interactive Demonstration

A biology teacher introduces the website during a lesson. Students follow along on desktops or laptops.

Self-Guided Learning

An individual who want to learn more about sleep health explores the experience at home on their personal desktops or laptops.

III. Research

This section covers the research on the topic, competitors,
and current design trends that influence the design.

Research

Topic Contents

The primary content displayed on the website will be about...

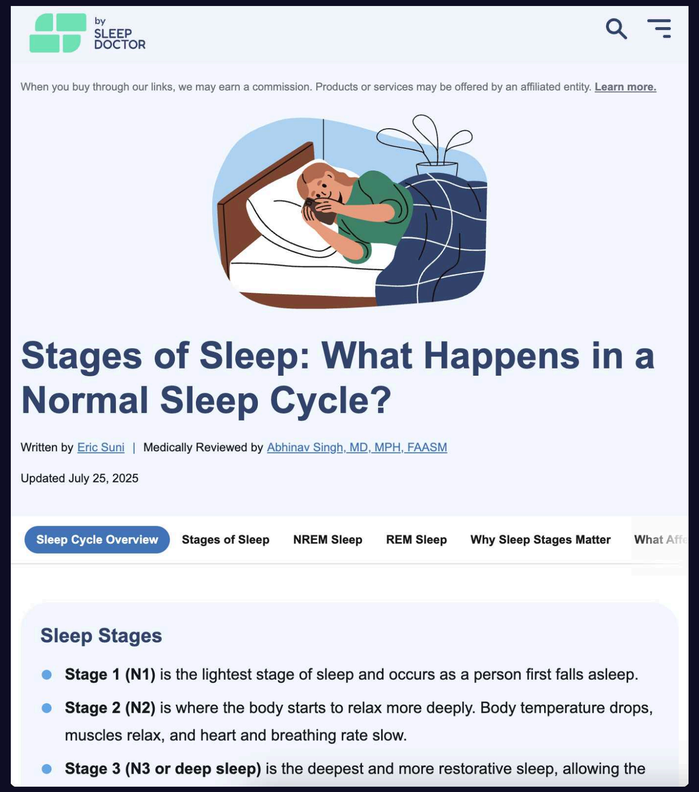
- **Physiological metrics** (heart rate, breathing changes)
- **Brain wave** frequency ranges (Beta, Alpha, Theta, Delta)
- **Hormonal cycles** (melatonin, cortisol, growth hormone)

Research

Competitive Analysis

Sleep Cycle Websites

Highly educational but often simplifies sleep sciences or doesn’t go into how sleep or each stage affects the body.



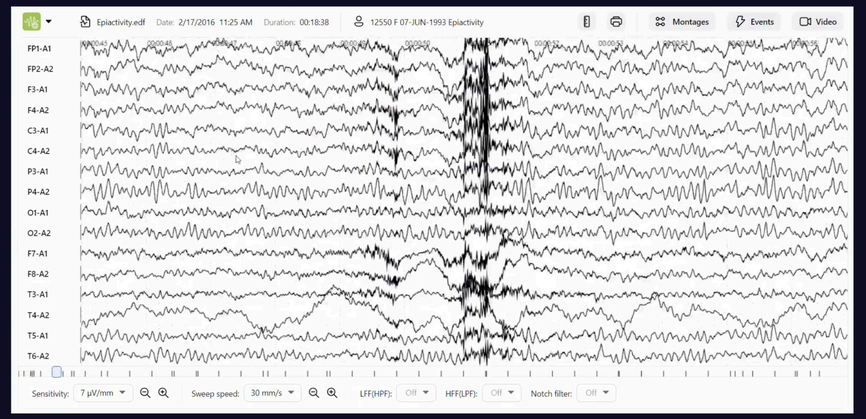
(eg. Stages of Sleep)

Research

Competitive Analysis

Sleep Cycle Simulations

Highly interactive but very technical with a high learning curve. Educational but typically very specialized.



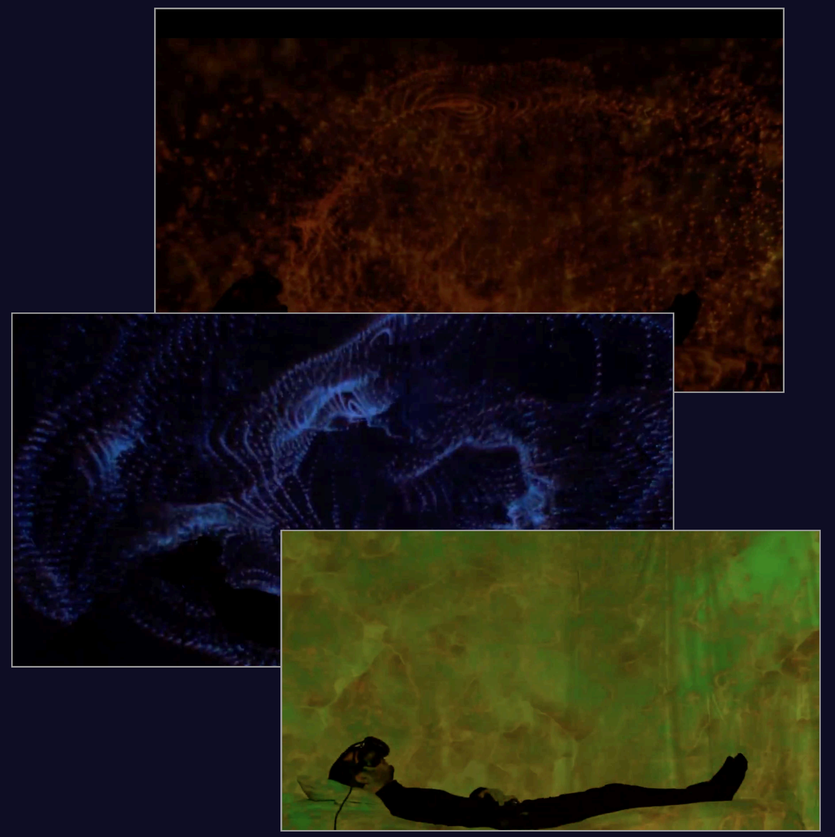
eg. Web EEG

Research

Competitive Analysis

Experiential Projects

Highly engaging but often VR/installation-based and not widely accessible. Immersive but typically not very educational.



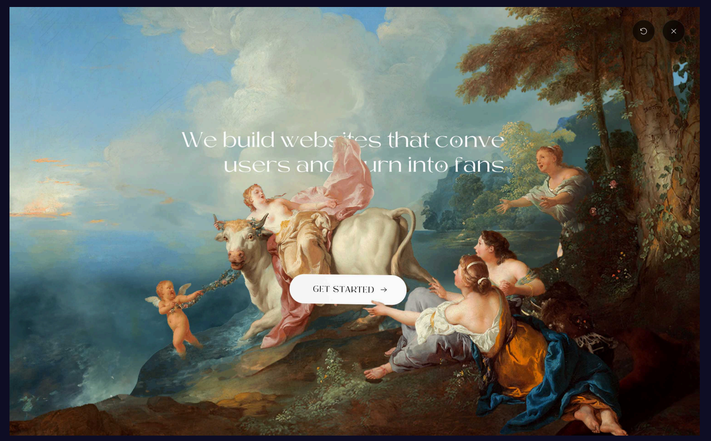
eg. Inter-Dream

Research

Design Trends

Parallax/Layered Environments

Layers move at different speeds to create depth and immersion. This makes interfaces feel more dynamic and immersive.





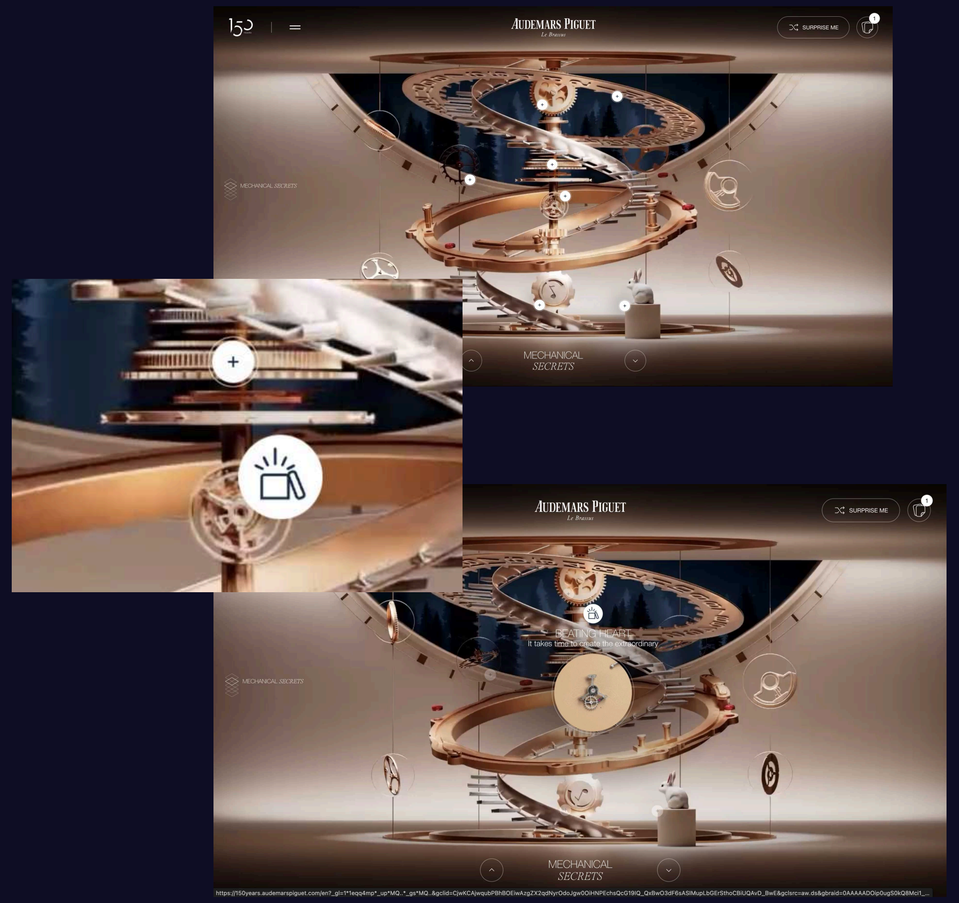
https://app.spline.design/community/file/8cf76a7f-428e-4bf7-af91-720754f07596

Research

Design Trends

Hotspot-based Disclosure

Information appears when users interact with specific areas. This keeps layouts clean and encourages exploration.



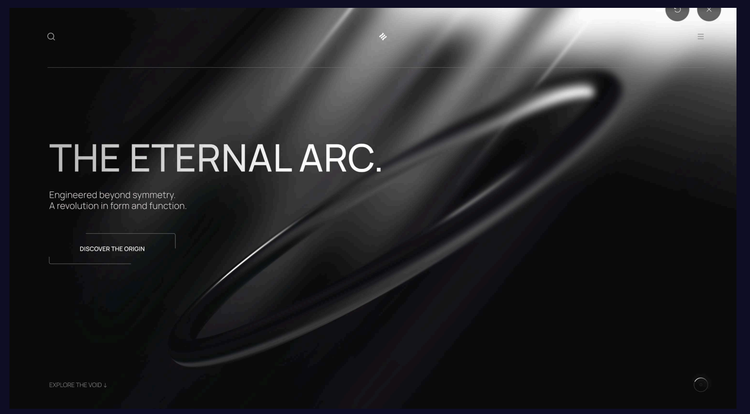
https://app.spline.design/community/file/8cf76a7f-428e-4bf7-af91-720754f07596

Research

Design Trends

Cursor Reactions

The cursor triggers visual feedback as users move or interact. This creates responsiveness, engagement, and delight.





https://app.spline.design/community/file/8cf76a7f-428e-4bf7-af91-720754f07596

IV. Planning

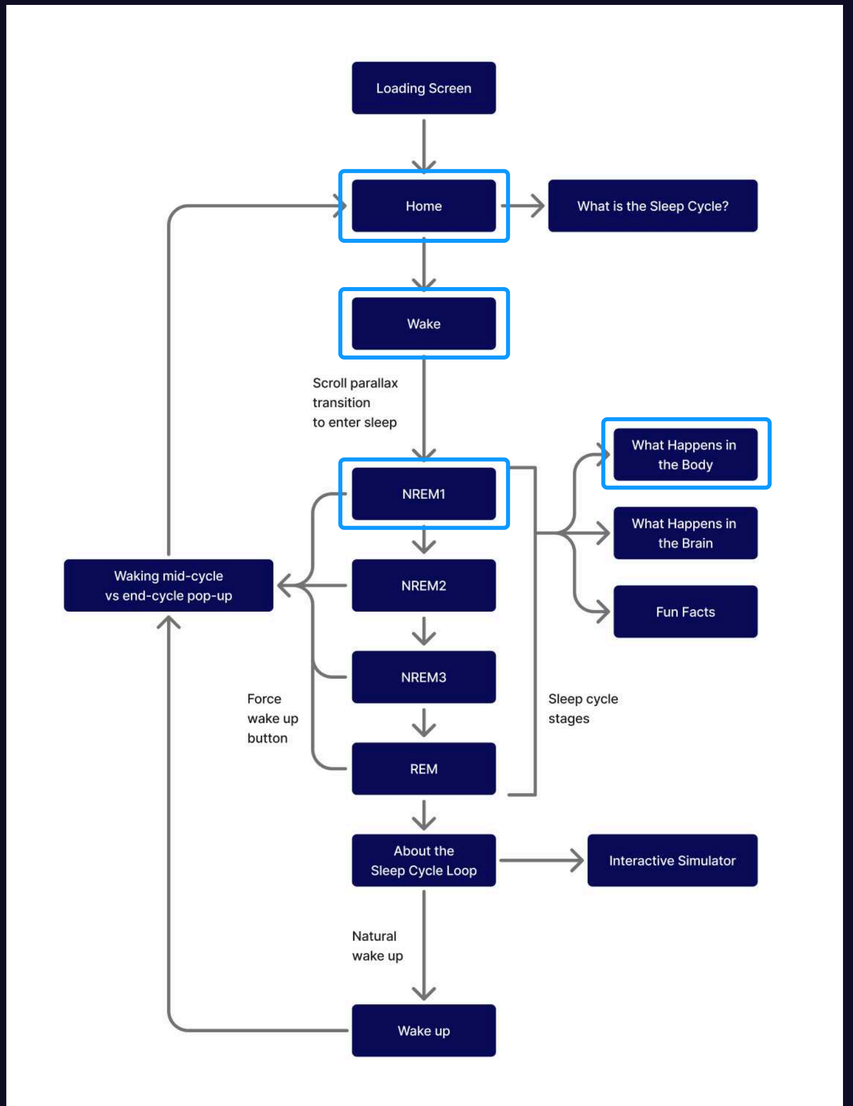
This section covers the site map, concept sketches, and several iterations of wireframes

Planning

Site Map

The sitemap indicates the different sections and possible interactions. Instead of separate pages, each section is revealed through scrolling.

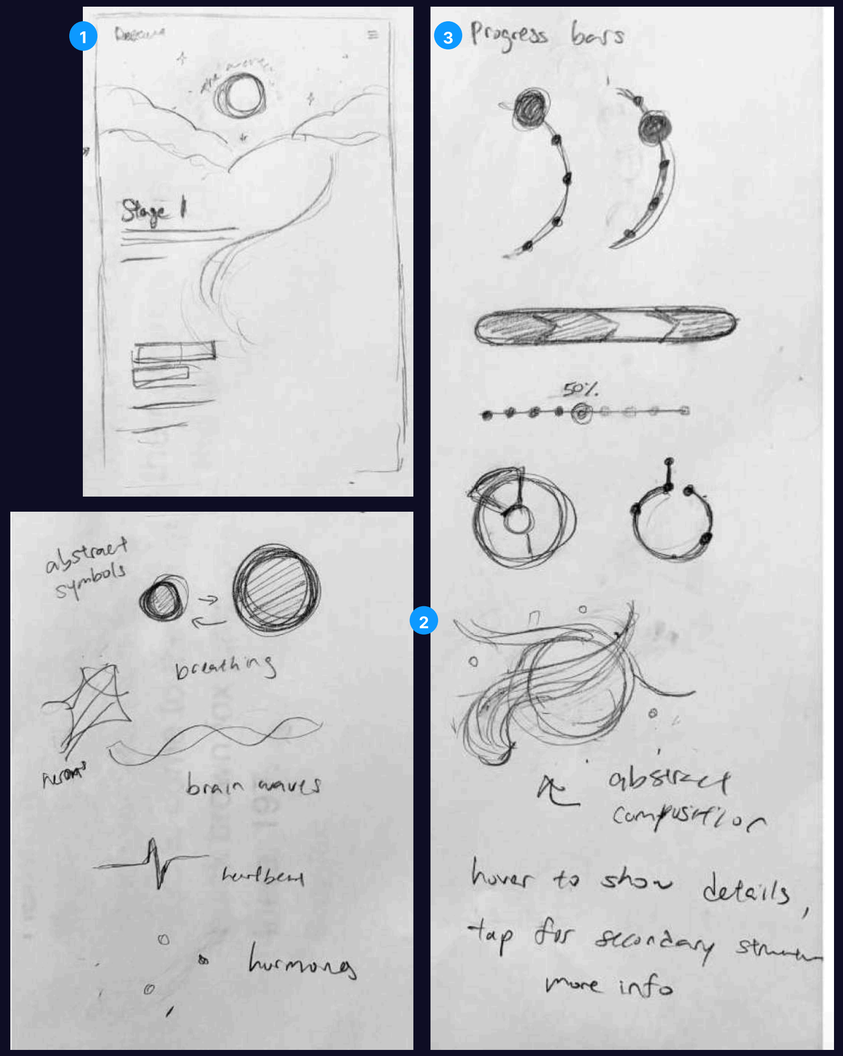
The prototype will focus on the highlighted sections.



Planning

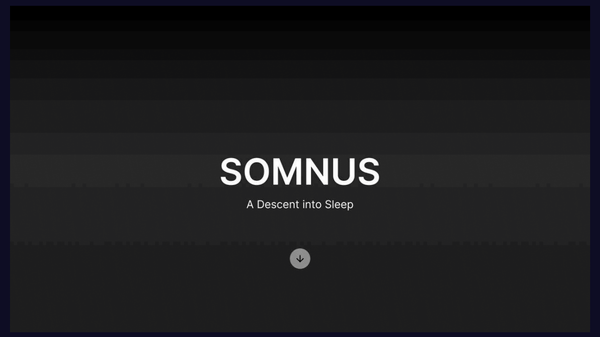
Concept Sketches

1. General Website Layout
2. Abstract Representations
3. Progress Bars



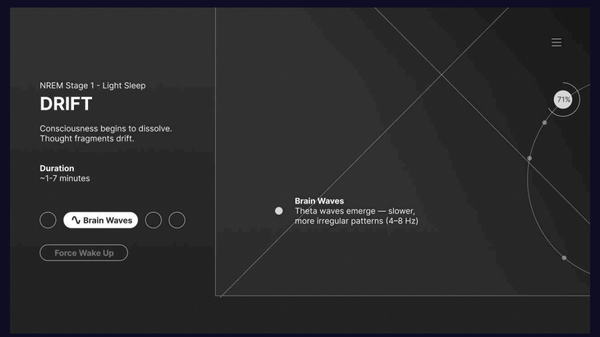
Planning

Initial wireframes









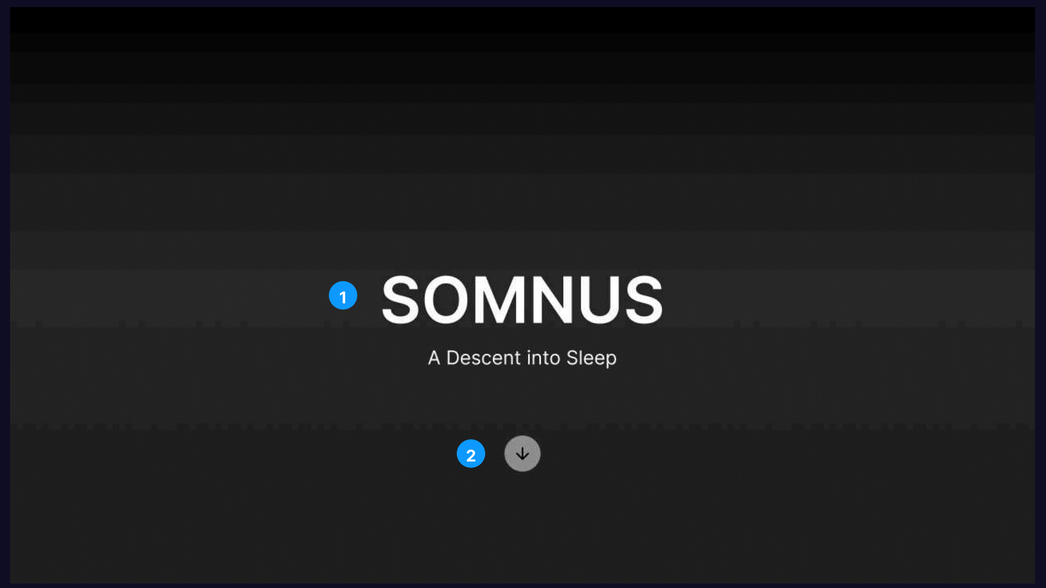


Planning

Landing Screen

1. Title
Somnus is the latin word for sleep, dream, drowsiness, and night. Since it is not a well-known word. It is accompanied by a subtitle.

2. Scroll Down Icon
An icon is used signal the user to scroll down.



Planning

Awake Screen

1. Guiding Question
Introduces the main question being answered on the website.

2. Scroll Down Icon
An icon is used signal the user to scroll down. In this screen, it is used to represent the moon.

3. Clouds
The clouds will be used to create the transition between the awake state and the sleep cycle stages using parallax animations.

4. Menu
Menu button is available but not prominent as the website is best experienced through interaction.



Planning

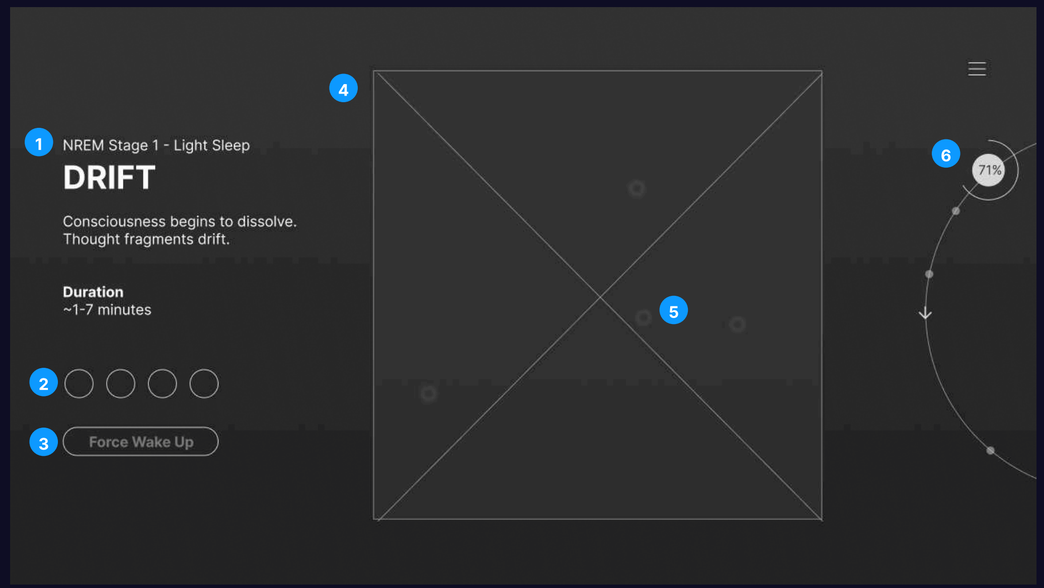
NREM 1 Screen

1. Content
Contains the stage of sleep, an descriptor of the feeling as the title, a short description, and the duration of the stage.

2. Hidden Information
The information is hidden until the user interacts with it on the 3D model or by clicking the button to reveal it.

3. Force Wake Up
Starts an interaction that lets the user to wake up from any stage of sleep and learn about it.

4. 3D Interactive Model
The main source of interaction and learning. The user can hover and interact with the abstract 3D composition to learn more.



Planning

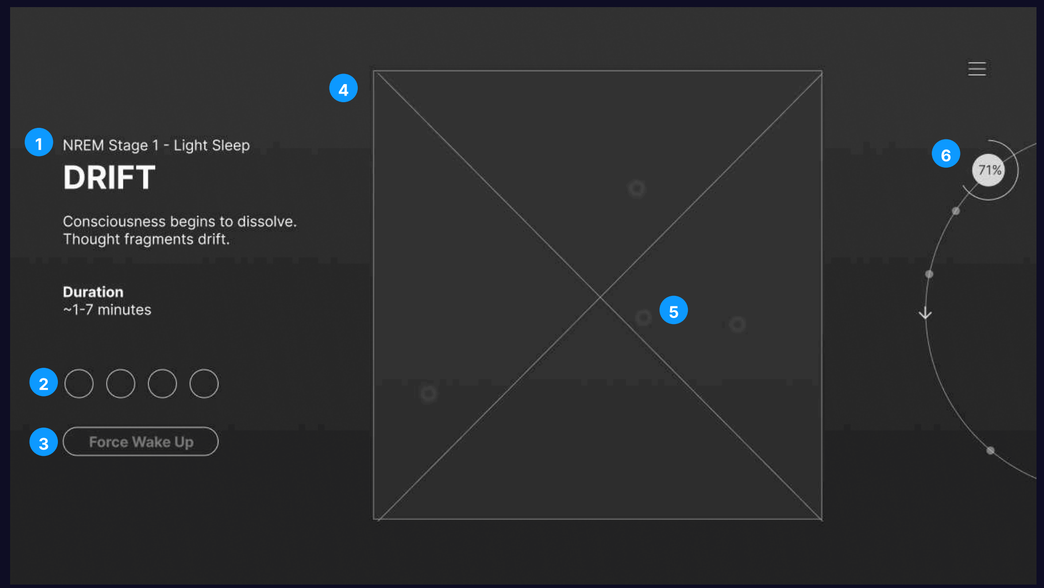
NREM 1 Screen (cont.)

5. Hidden Interactions

The dots will be hidden but very subtly visible to help the user find its location.

6. Progress Bar

The enlarged dot represents the stage the user is currently at. The percentage inside and the outline around it represents the percentage of the stage that has been explored.



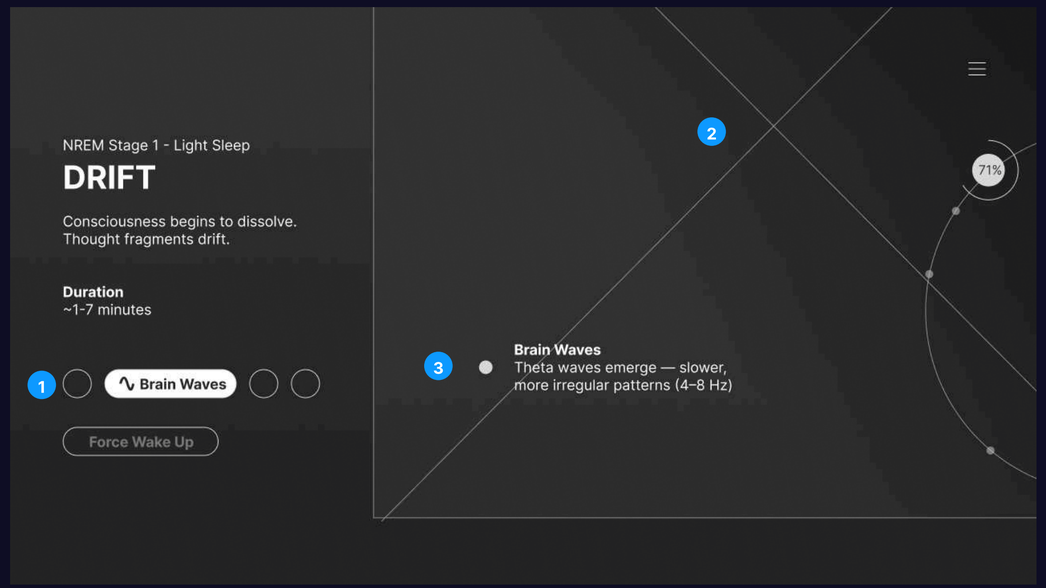
Planning

Secondary Screen

1. Found Interactions
When an interaction is found by the user, the button becomes permanently filled with an icon. While it is being interacted with, the button will expand to show the name.

2. Zoomed 3D Model
While an interaction is clicked on, the 3D model will zoom in and focus on that specific area.

3. Information
When interacted with, the dot shows a title, information, and a graph or chat if it fits the content.



Planning

Force Wake Up Screen

1. Initiated Forced Wake Up

Once clicked and initiated, the button becomes filled and interaction starts.

2. Forced Wake Up Interaction

The progress bar and information about waking up during the stage shifts to the middle while the 3D model fades out. It now contains a button that the user can use to wake up through an animated transition back to the home screen. Different stages will have different animations to show the difference in difficulty of waking up at certain stages.

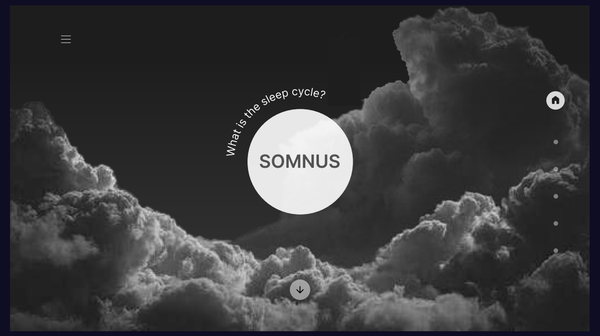
3. Exiting the Interaction

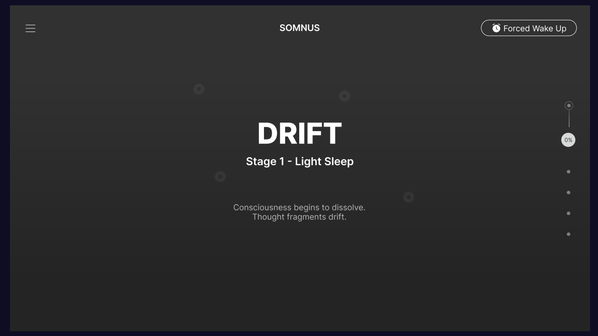
Users can click the arrow or click the Forced Wake Up button again to cancel the interaction



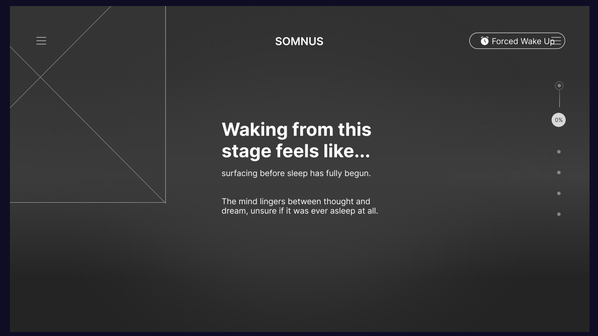
Planning

Refined wireframes









Planning

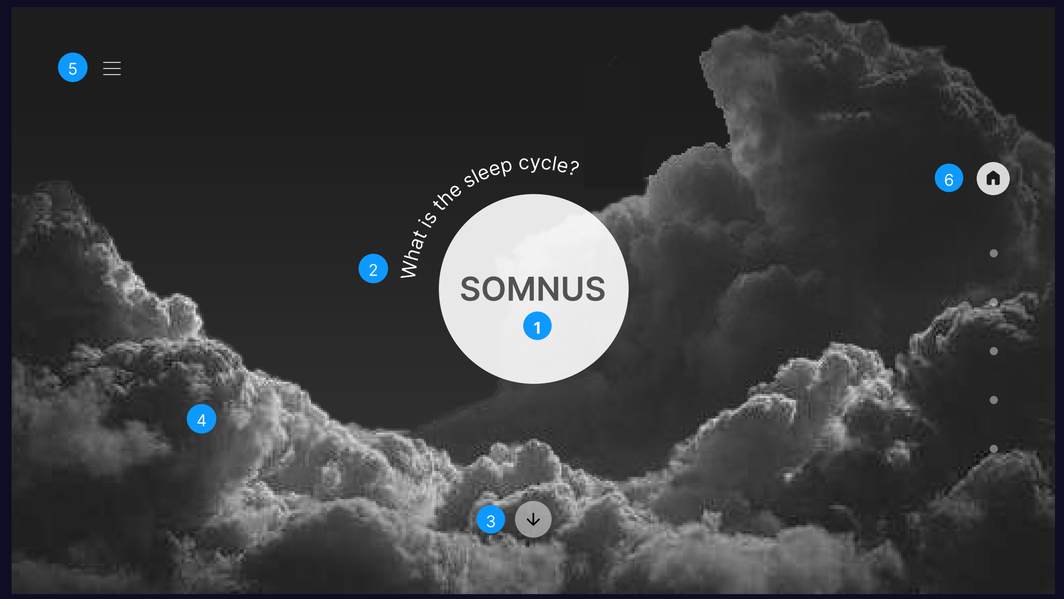
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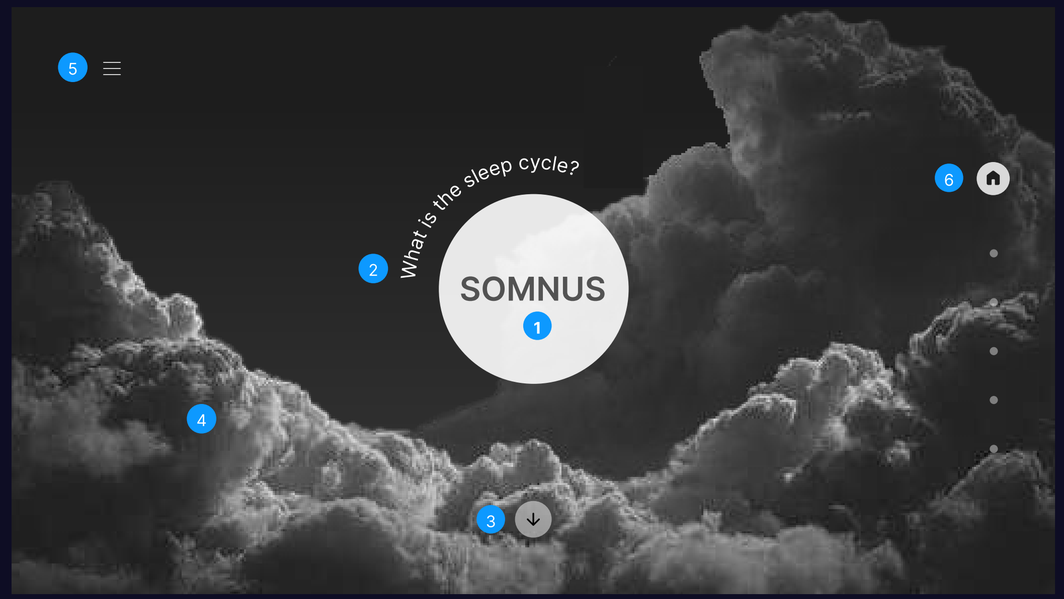


Planning

Awake Screen (cont.)

5. Menu
Menu button is available but not prominent as the website is best experienced through interaction.

6. Progress Bar
The enlarged dot represents the stage the user is currently at. The percentage inside and the outline around it represents the percentage of the stage that has been explored.



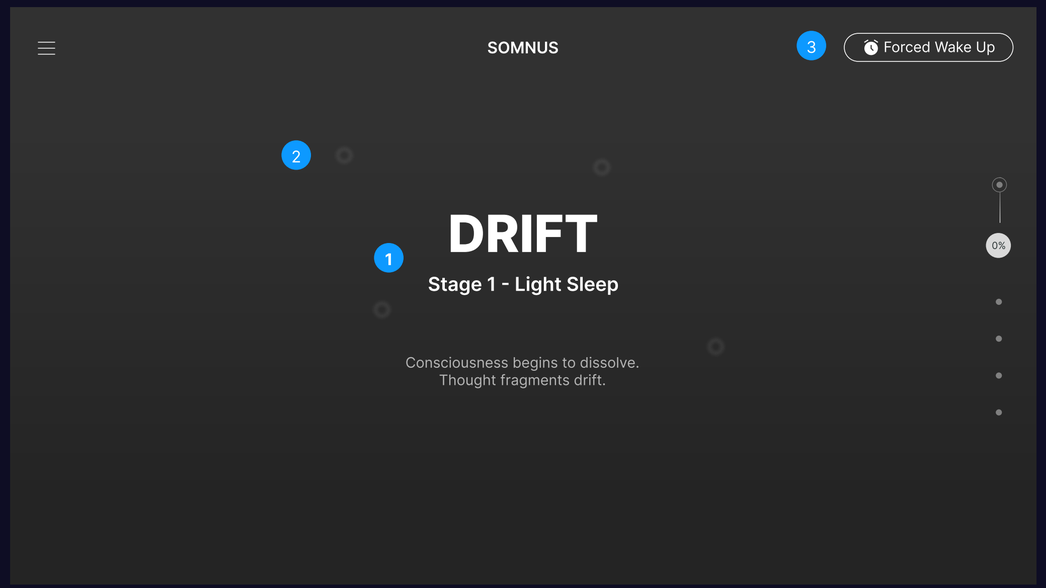
Planning

NREM 1 Screen

1. Content
Contains the stage of sleep, an descriptor of the feeling as the title, and a short description.

2. Hotspots
The information is hidden until the user interacts with it on the 3D model or by clicking the button to reveal it.

3. Force Wake Up
Starts an interaction that lets the user to wake up from any stage of sleep and learn about it.



Planning

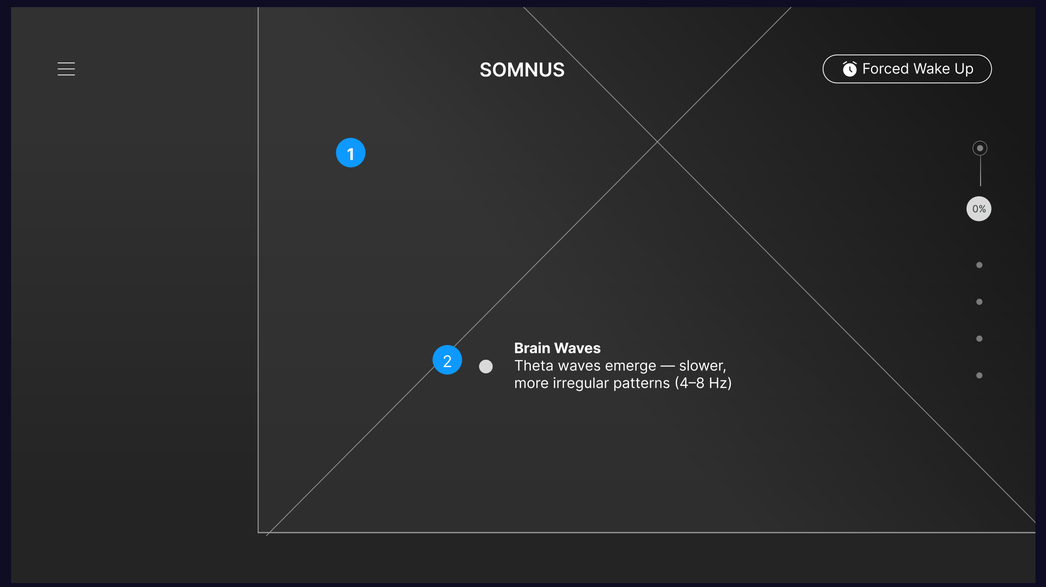
Secondary Screen

1. Zoomed 3D Model

While an interaction is clicked on, the 3D model will zoom in and focus on that specific area.

2. Information

When interacted with, the dot shows a title, information, and a graph or chat if it fits the content.



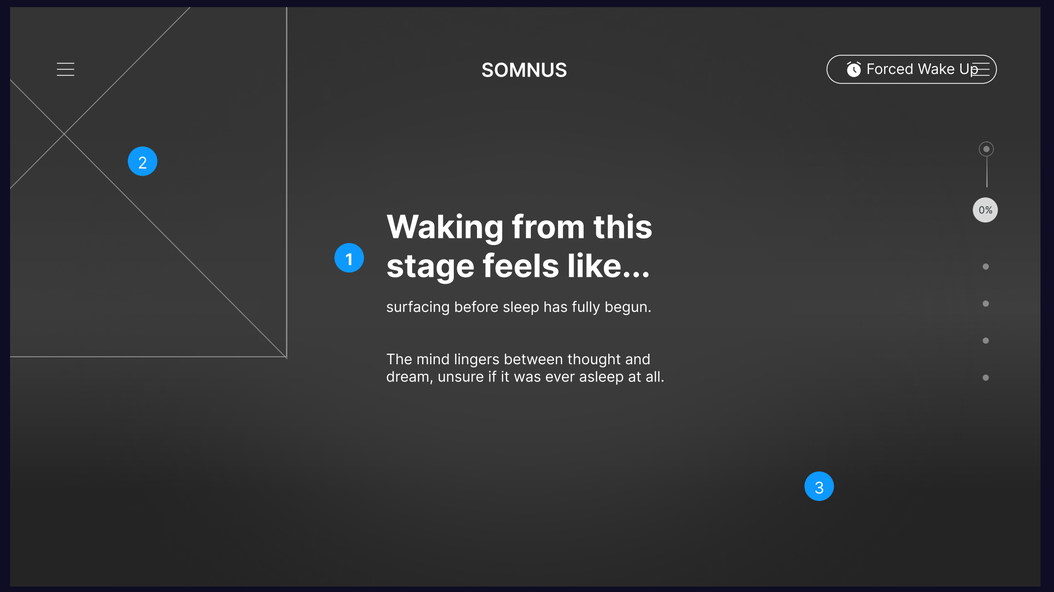
Planning

Wake Up Screen

1. Waking Content
Describes what waking up at the stage feels like.

2. 3D Model
The 3D model shifts to the side and becomes less prominent.

3. Animation
The user reads the animated text, the screen blanks out and transitions to the home screen. The transition depends on how difficult it is to wake up at the stage. Then, the website automatically scrolls down to where the user left off.



V. Visual Development

This section covers the inspirations, mood/style, and the visual design development of the interactive website.

Visual Development

Moodboard I

Luminous Drift

Flowy black-and-white photography accented by colored light, with visible light rays and floating particles creating depth and atmosphere.

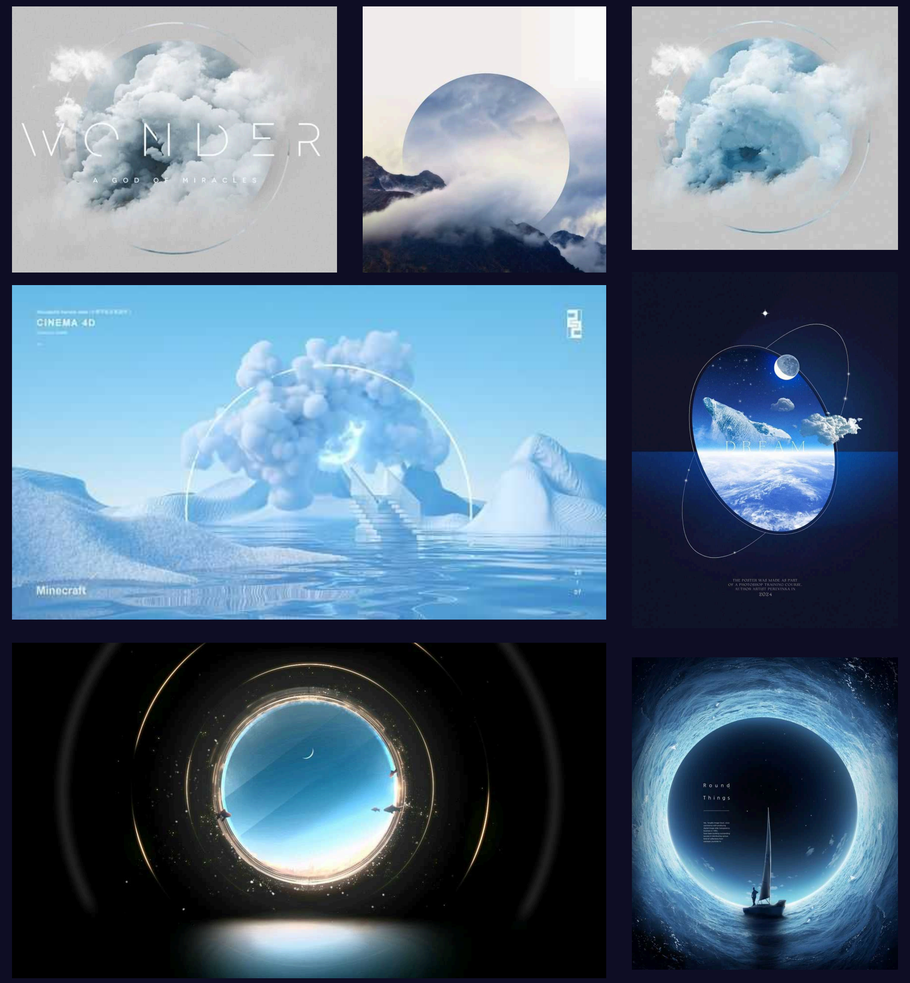


Visual Development

Moodboard II

Geometric Surrealism

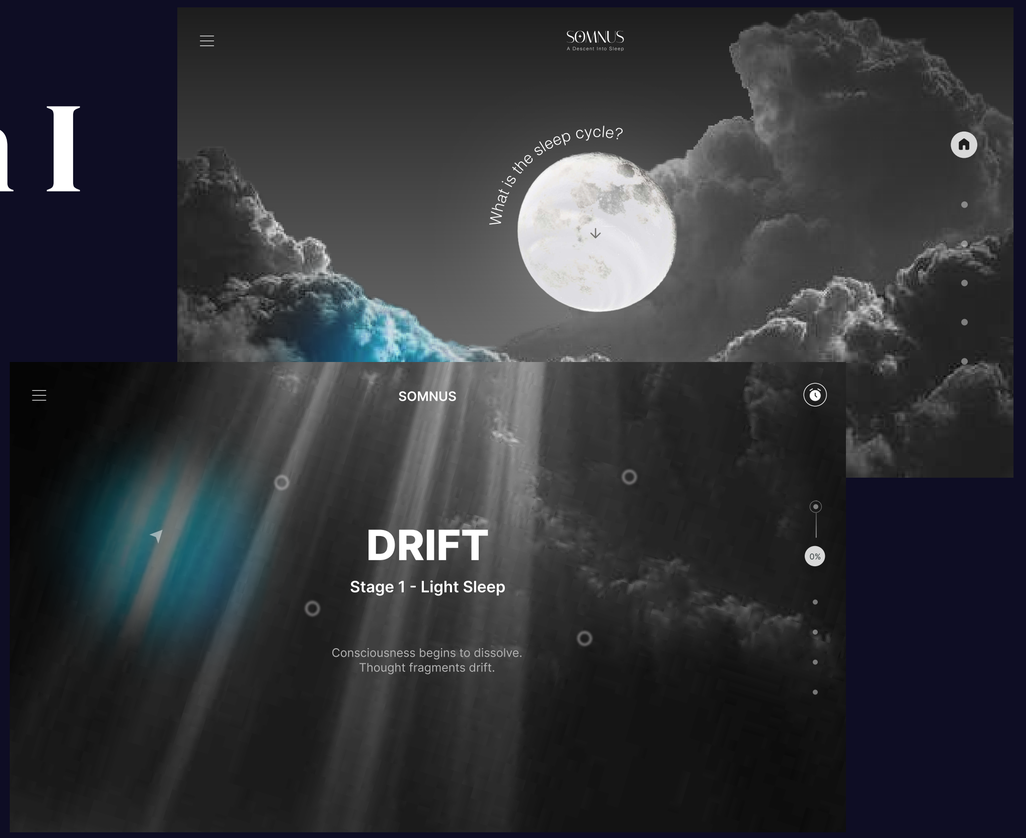
Surreal, geometric forms in a mono or duo-chromatic palette, defined by a clean, digitally constructed “Photoshop” look.



Visual Development

Visual Composition I

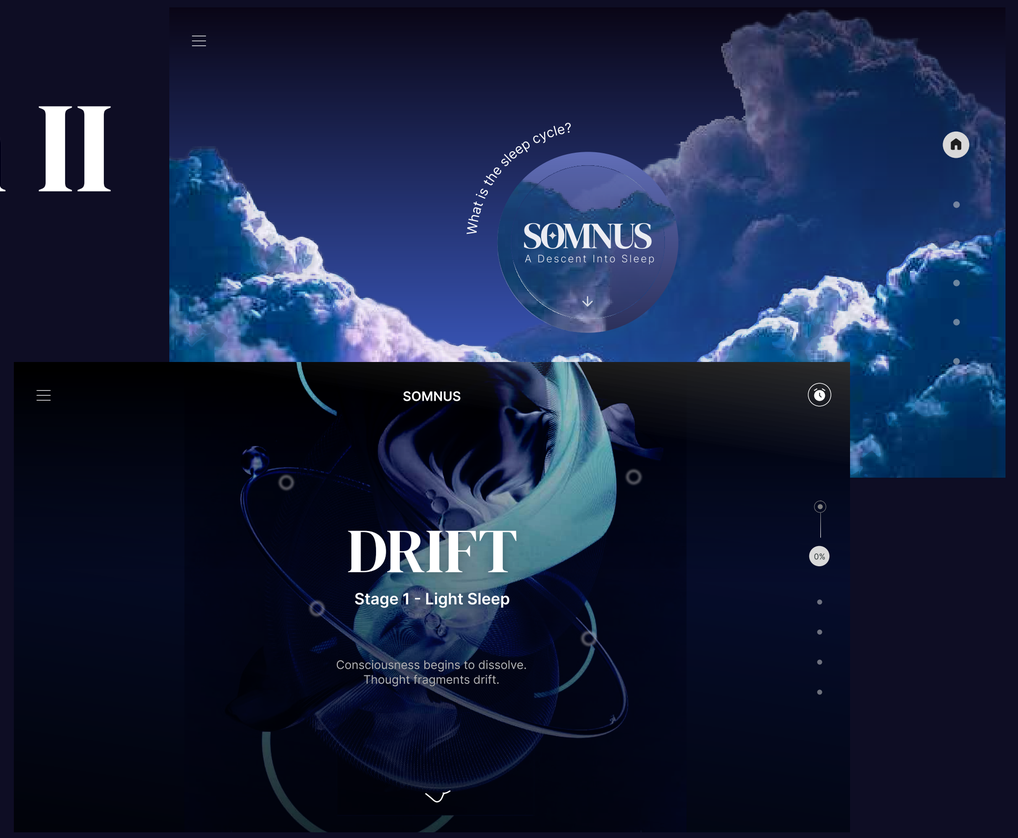
Applying the Luminous Drift moodboard to create a black and white atmospheric scene with a colored light that follows the cursor.



Visual Development

Visual Composition II

Applying the Geometric Surrealism moodboard to create an abstract and geometric surreal environment.



Visual Development

Further Development

Refined the logo, reduced progress navigation size, moved 3d model interaction to within hotspot interaction.



VI. Final Designs

This section covers the style guide, assets, final design screens, and the viewable prototype.

Final Designs

Style Guide

DM Serif Display Regular

Inter Semibold

Inter Regular

Inter Light

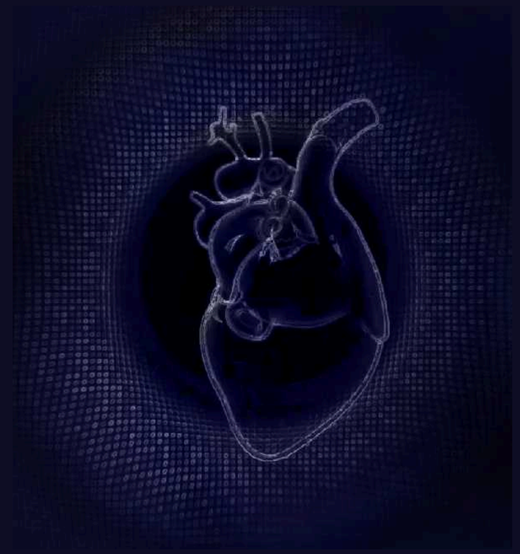
Typography

Colors

A Descent into Sleep

SOMNUS

Logo









Assets

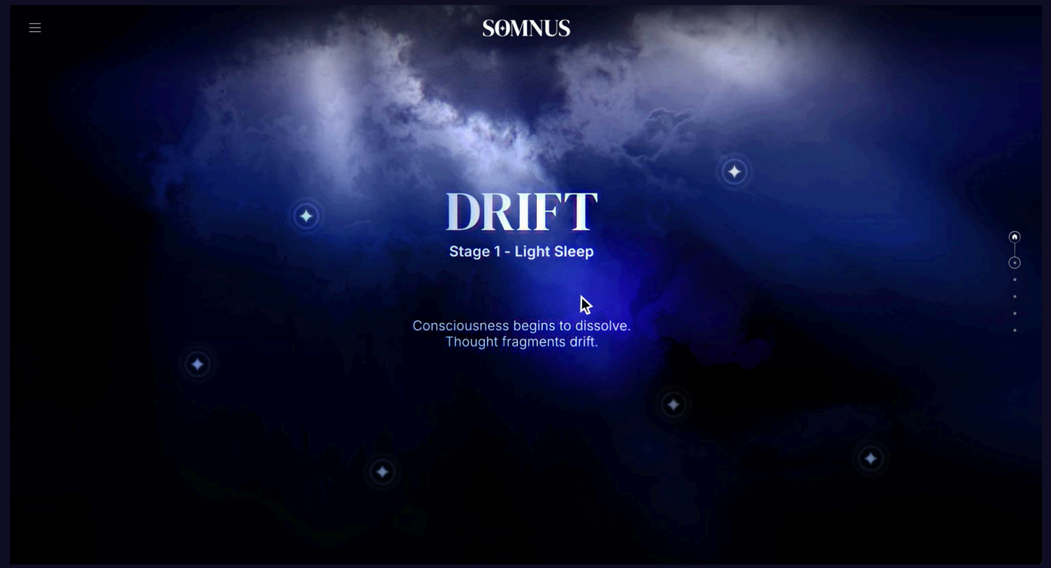
Final Designs

Awake Screen (Home)



Final Designs

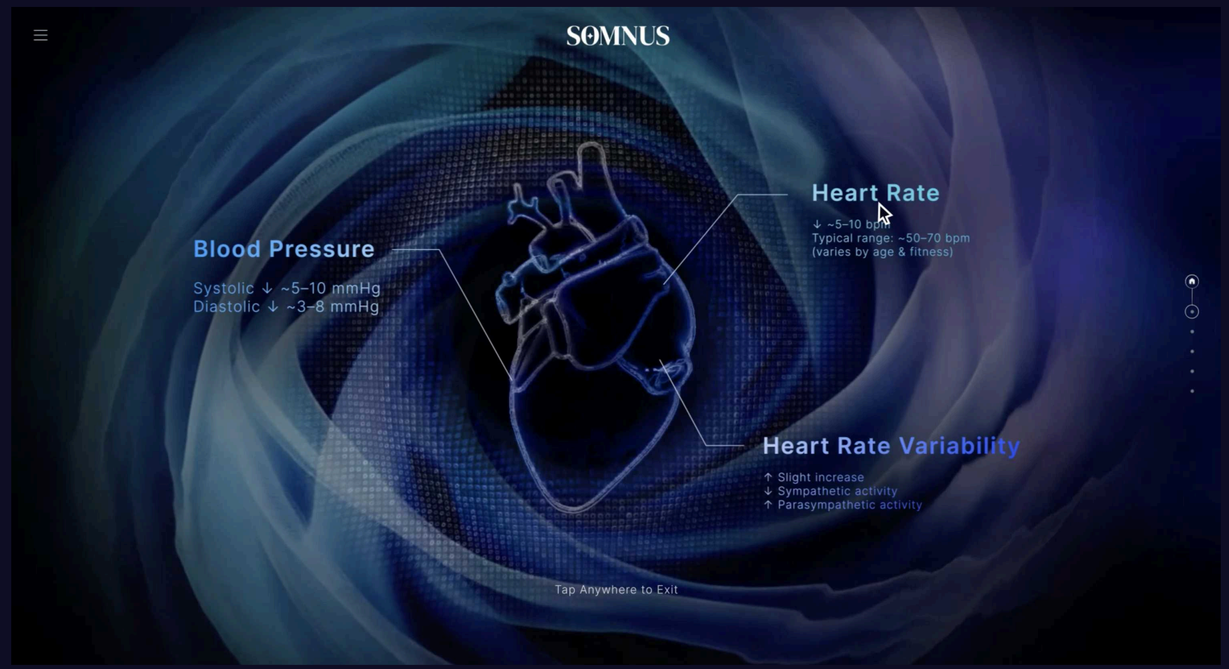
NREM Stage I Screen





Final Designs

Secondary Screen



Final Designs

Prototype

- Parallax mouse tracking and scroll
- Interactive hotspot hover state
- Progressive disclosure of information



The prototype is available on Vimeo.

VII. Conclusion

This section concludes the current project by covering the takeaways, plans for future development, and citation.

Visual Development

Takeaways

Strong research grounded the project and revealed opportunities to **differentiate through engaging motion and interactivity**.

By using an animated prototype with recorded interactions, I was able to **communicate ideas and interactions effectively without being constrained by technical limitations.**

Visual Development

Future Development

If I have more time, I would like to...

- Make the secondary page more easily **digestible or interactive**
- Animate **all stages and transition** to home (natural wake up)
- Animate the **forced wake up transitions** that differ based on the stage of sleep

Visual Development

Citations

- National Sleep Foundation. “Stages of Sleep.”
- Inter-Dream
- Audemars Piguet
- Parallax Interactive Scene
- Cursor Light Interaction
- Harvard Medical School Sleep Lab. “Sleep Stages and Brain Activity.”
- Corn. Revolutionized.